

A Fixed-Shift Arithmetic Transfer Gap: From Literal Completion Certificates to Zero-Frequency Flatness

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Abstract

The distinguished zero determinant frequency in a fixed-shift prime–Möbius packet is controlled by a ratio, not by an amplitude estimate alone. If $a_X = (A_{C_X}(n))_n$ is the literal determinant coefficient, the required quantity is

$$\mathfrak{F}_{h_0}(X) = \frac{|\sum_n A_{C_X}(n)|^2}{\sum_n |A_{C_X}(n)|^2}.$$

Earlier work isolated the threshold $\mathfrak{F}_{h_0}(X) \leq X^{1/400+o(1)}$ and an amplitude target $|\sum_n A_{C_X}(n)| \ll X^{o(1)} Q_X^2$, but did not supply the matching lower bound for the denominator.

This paper gives an exact transfer architecture for that missing denominator. A represented synthesis $\mathcal{A}_X$, a literal missing synthesis $\mathcal{M}_X$, and their reassembly $\mathcal{B}_X = \mathcal{A}_X + \mathcal{M}_X$ yield coefficient-specific and uniform energy survival bounds from a normalized missing-Gram Perron certificate. An explicit determinant-binning map $\mathcal{J}_X$, a verified restricted lower modulus s_X , and the dictionary defect $r_X = a_X - \mathcal{J}_X \mathcal{B}_X z_X$ then give

$$\|a_X\|_2^2 \geq \left(s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}} - \frac{\|r_X\|_2}{\sqrt{D_{\text{raw},X}}} \right)_+^2 D_{\text{raw},X}.$$

At the high-row scales $Q_X = X^{267/400+o(1)}$ and $J_X = X^{133/400+o(1)}$, an amplitude exponent η_Z and an energy-loss exponent λ_D give

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400+\lambda_D-2\eta_Z+o(1)}.$$

For this exponent transfer, the sharp recorded-bounds condition is $2\eta_Z \geq \lambda_D$, with genuine robustness only under strict inequality. It is not claimed to be necessary for the physical packet when the recorded exponents are nonoptimal.

Two sharp countermodels show why this bridge is not formal: unsigned support and energy do not determine the zero mode, and raw atomic mass can disappear completely inside one determinant bin. Accordingly, the results are exact L0 algebra and a conditional L1 arithmetic interface. They do not prove determinant-energy survival for the physical packet, a full-block cover, a fixed-shift Hardy–Littlewood asymptotic, twin primes, or a breach of the parity barrier.

Keywords: fixed shift; provenance completion; determinant binning; zero-frequency flatness; energy survival; Gram majorant; parity barrier.

Convention. Every physical statement retains one fixed nonzero shift h_0 , literal coefficients, complete native keys, exact post-aggregation support, and one global raw atomic normalizer. No averaged-shift, frozen-family, or support-only substitute is accepted as a fixed- h_0 arithmetic estimate.

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1 Introduction

Fix one nonzero integer shift h_0 . The high-row branch of the fixed-shift program reaches a finite determinant coefficient

$$a_X = (A_{C_X}(n))_{n \in \mathcal{N}_X}, \quad C_X = \lfloor J_X \rfloor, \tag{1}$$

formed after literal aggregation of a complete set of native prime–Möbius keys. Its zero mode and determinant energy are

$$Z_X := \sum_{n \in \mathcal{N}_X} A_{C_X}(n), \quad D_X := \sum_{n \in \mathcal{N}_X} |A_{C_X}(n)|^2. \quad (2)$$

The zero-frequency concentration is

$$\mathfrak{F}_{h_0}(X) := \begin{cases} |Z_X|^2/D_X, & D_X > 0, \\ 0, & D_X = 0. \end{cases} \quad (3)$$

The zero convention in the second line is harmless only because then $a_X = 0$ and hence $Z_X = 0$. Here “zero frequency” means the normalized determinant-DFT frequency $r = 0$ used in TPC-32. It is neither a generic nonzero determinant frequency nor any orbit, divisor, or transfer-operator zero mode.

TPC-32 proved a conditional transfer at

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)} : \quad (4)$$

the estimate

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400+o(1)} \quad (5)$$

delivers every strict saving below 133/400 for its complete matched raw cutoff shell [7]. TPC-33 isolated the sufficient amplitude estimate

$$|Z_X| \ll X^{o(1)} Q_X^2, \quad (6)$$

but explicitly did not prove a matching lower bound for D_X [8]. Indeed, (6) alone says nothing about (5) when the denominator may be small.

The purpose of this paper is to identify exactly what a completion certificate must additionally prove before it controls D_X . The finite completion, sparse-incidence, and scalar Perron layers that motivate this interface were developed in [9–11]; none of those layers by itself identifies the determinant coefficient. The relevant normalized statistic is

$$\mathfrak{D}_{h_0}(X) := \frac{D_X}{Q_X^3 J_X^2}. \quad (7)$$

At the scales (4), the denominator $Q_X^3 J_X^2$ is the energy scale that pairs with (6). The needed zero-loss statement is

$$\mathfrak{D}_{h_0}(X) \geq X^{-o(1)}. \quad (8)$$

No such physical estimate is proved here.

1.1 The three interfaces

There are three logically different maps:

$$\text{native coefficients} \xrightarrow{\mathcal{A}_X + \mathcal{M}_X} \text{pre-bin atoms} \xrightarrow{\mathcal{J}_X} \text{determinant bins} \xrightarrow{\mathbf{1}^*} \text{zero mode}.$$

The first arrow is a completion/reassembly problem. The second can lose energy through collisions in a determinant fiber. The third is the arithmetic signed sum. Success at one arrow does not imply success at either later arrow.

For the first arrow we prove a missing-Gram majorant theorem. If $\mathcal{B}_X = \mathcal{A}_X + \mathcal{M}_X$, then for every coefficient vector z with $\mathcal{A}_X z \neq 0$,

$$\left(1 - \sqrt{\Xi_X(z)}\right)_+^2 \leq \frac{\|\mathcal{B}_X z\|_2^2}{\|\mathcal{A}_X z\|_2^2} \leq \left(1 + \sqrt{\Xi_X(z)}\right)^2, \quad (9)$$

where

$$\Xi_X(z) := \widehat{\kappa}_X \frac{\langle D_{\mathcal{M}, X} z, z \rangle}{\|\mathcal{A}_X z\|_2^2}, \quad \widehat{\kappa}_X = 1 + \rho(P_X). \quad (10)$$

Here $D_{\mathcal{M},X}$ is the exact diagonal of the missing Gram and P_X is an entrywise nonnegative majorant of its normalized off-diagonal part.

For the second arrow we require a literal binning map, not an analogy. If s_X is a verified lower modulus of $\mathcal{J}_X$ on the actual reassembled subspace and

$$r_X = a_X - \mathcal{J}_X \mathcal{B}_X z_X,$$

then the lower bound displayed in the abstract follows. The residual r_X records every mismatch between the completion carrier and the physical determinant coefficient.

For the final arrow we prove the exact exponent arithmetic. If

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2, \quad D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2, \quad (11)$$

then

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400 + \lambda_D - 2\eta_Z + o(1)}. \quad (12)$$

Consequently these recorded bounds recover the TPC-32 threshold whenever $2\eta_Z \geq \lambda_D$, and this condition is sharp for the two bounds alone. The current amplitude target has $\eta_Z = 0$; within this transfer ledger it therefore permits no positive certified energy loss. Such a loss cannot be absorbed unless it is offset by a new amplitude saving.

1.2 Level ledger

The level labels used throughout are:

- **L0:** exact finite-dimensional algebra, including Gram majorization, reassembly inequalities, and determinant-binning identities;
- **L1:** a verified literal interface from those identities to the actual fixed- h_0 coefficient, together with asymptotic bounds for every certificate statistic;
- **L2:** a new arithmetic estimate for the actual fixed- h_0 packet, and then a separately verified global block cover.

This paper proves L0 theorems and states auditable L1 implications. It proves no new L2 cancellation estimate. In particular, none of the following can be inferred from the results below: a fixed-shift prime-pair asymptotic, positivity at $h_0 = 2$, twin primes, or a breach of the sieve parity barrier [1].

1.3 Organization

[Section 2](#) defines the physical packet and its global normalization. [Section 3](#) proves the coefficient-specific Gram bound, and [section 4](#) extracts a uniform survival certificate. [Section 5](#) adds the determinant dictionary. [Section 6](#) gives the exponent theorem. [Section 7](#) supplies sharp no-go models. [Section 8](#) states the conditional global transfer, and [section 9](#) gives the archive and stop protocol.

2 The literal fixed-shift packet

All spaces in the finite algebra below are complex Hilbert spaces with their standard inner products, linear in the first variable. The same statements hold over real spaces. The index X is suppressed when no ambiguity is possible.

Definition 2.1 (Complete native packet). A complete native packet at scale X consists of:

- (P1) one fixed nonzero integer h_0 , not averaged over a family;
- (P2) a finite native key set $\mathcal{K}_X$, including all row, orbit, content, divisor, orientation, source, and endpoint labels used by the construction;
- (P3) a coefficient space $\mathcal{H}_X = \mathbb{C}^{\mathcal{K}_X}$ and the literal physical coefficient $z_X \in \mathcal{H}_X$;

(P4) a pre-bin key set $\mathcal{Y}_X$ and literal maps

$$\mathcal{A}_X, \mathcal{M}_X : \mathcal{H}_X \longrightarrow \mathbb{C}^{\mathcal{Y}_X}, \quad \mathcal{B}_X := \mathcal{A}_X + \mathcal{M}_X;$$

(P5) a determinant key set $\mathcal{N}_X$, an exact determinant label $\nu_X : \mathcal{Y}_X \rightarrow \mathcal{N}_X$, and all literal scalar multipliers attached to the aggregation;

(P6) the actual post-aggregation coefficient $a_X = (A_{C_X}(n))_{n \in \mathcal{N}_X}$, with exact zeros retained until the aggregation has been performed;

(P7) one global raw atomic energy $D_{\text{raw},X} > 0$, computed before block deletion, renormalization, thresholding, or selection.

The words “represented” and “missing” describe provenance, not sign. Thus $\mathcal{M}_X$ already includes its literal signs and phases; reassembly is $\mathcal{A}_X + \mathcal{M}_X$, never an unsigned union of supports. Similarly, $D_{\text{raw},X}$ is not recomputed on the represented support. Per-component normalizers can conceal an arbitrarily large deleted mass and are therefore inadmissible for the physical transfer.

Definition 2.2 (Exact determinant binning). Let $\tau_X(y) \in \mathbb{C}$ be the archived multiplier on a pre-bin key $y \in \mathcal{Y}_X$. Define

$$(\mathcal{J}_X u)(n) := \sum_{\substack{y \in \mathcal{Y}_X \\ \nu_X(y)=n}} \tau_X(y) u(y), \quad n \in \mathcal{N}_X. \quad (13)$$

The sum is taken over every native pre-bin key in the fiber. The dictionary residual is

$$r_X := a_X - \mathcal{J}_X \mathcal{B}_X z_X. \quad (14)$$

If the proposed carrier is literally the physical coefficient, then $r_X = 0$ by exact finite arithmetic. If only a model, restriction, or surrogate carrier has been constructed, (14) does not vanish automatically. The residual is the correct location for that discrepancy.

Remark 2.3 (Actual support). The support of a synthesis map is its exact nonzero support after literal coefficient aggregation. A geometrically available edge, a thresholded numerical entry, or a union of possible carriers is not an actual edge. In particular, a spectral radius computed on an ambient support is not the Perron constant used below unless it is proved to majorize the normalized Gram on the actual active coordinates.

Remark 2.4 (No change of target). The completion maps take values in the pre-bin space $\mathbb{C}^{\mathcal{Y}_X}$, while the arithmetic coefficient lives in $\mathbb{C}^{\mathcal{N}_X}$. An observation on another quotient, a four-corner trace, or a common-target gauge does not define $\mathcal{J}_X$. The determinant map and every aggregation multiplier must be constructed explicitly.

For reference, write

$$Z_X = \langle a_X, \mathbf{1} \rangle, \quad D_X = \|a_X\|_2^2, \quad \mathfrak{D}_{h_0}(X) = \frac{D_X}{Q_X^3 J_X^2}. \quad (15)$$

These are physical, post-aggregation statistics. None is replaced by the corresponding statistic of $\mathcal{A}_X z_X$, $\mathcal{B}_X z_X$, or an unsigned majorant.

3 Completion-to-survival from the missing Gram

We first prove a finite-dimensional theorem independent of arithmetic. Let K, Y be finite index sets, let $\mathcal{A}, \mathcal{M} : \mathbb{C}^K \rightarrow \mathbb{C}^Y$ be linear maps, and put $\mathcal{B} = \mathcal{A} + \mathcal{M}$. The missing Gram is

$$G_{\mathcal{M}} := \mathcal{M}^* \mathcal{M}. \quad (16)$$

We use only standard Hermitian and nonnegative-matrix comparison principles [2]. Let

$$d_j = (G_{\mathcal{M}})_{jj} = \|\mathcal{M} e_j\|^2.$$

If $d_j = 0$, positivity of $G_{\mathcal{M}}$ gives $(G_{\mathcal{M}})_{ij} = 0$ for every i , so that coordinate contributes nothing to $\mathcal{M}$. We therefore restrict the Gram normalization to the exact active set

$$K^+ := \{j : d_j > 0\} \quad (17)$$

and extend all resulting quadratic inequalities by zero on its complement.

Define the full diagonal operator

$$D_{\mathcal{M}} := \text{diag}(d_j : j \in K) \quad \text{on } \mathbb{C}^K. \quad (18)$$

Let $G_{\mathcal{M}}^+$ and $D_{\mathcal{M}}^+$ be the principal restrictions of $G_{\mathcal{M}}$ and $D_{\mathcal{M}}$ to K^+ , and set

$$R_{\mathcal{M}} := (D_{\mathcal{M}}^+)^{-1/2}(G_{\mathcal{M}}^+ - D_{\mathcal{M}}^+)(D_{\mathcal{M}}^+)^{-1/2}. \quad (19)$$

Thus $R_{\mathcal{M}}$ acts on $\mathbb{C}^{K^+}$ and is Hermitian with zero diagonal, while $\langle D_{\mathcal{M}}z, z \rangle$ below is the full zero-extended quadratic form on $\mathbb{C}^K$.

Definition 3.1 (Verified Perron majorant). A verified Perron majorant is a real symmetric nonnegative matrix P on K^+ such that

$$|(R_{\mathcal{M}})_{ij}| \leq P_{ij} \quad (i, j \in K^+). \quad (20)$$

We put

$$\hat{\kappa} := 1 + \rho(P). \quad (21)$$

If $K^+ = \emptyset$, P and $R_{\mathcal{M}}$ are the unique empty matrices and we set $\rho(P) = 0$, so $\hat{\kappa} = 1$.

Lemma 3.2 (Missing-energy majorization). For every $z \in \mathbb{C}^K$,

$$\|\mathcal{M}z\|^2 \leq \hat{\kappa} \langle D_{\mathcal{M}}z, z \rangle. \quad (22)$$

Proof. If $K^+ = \emptyset$, then $\mathcal{M} = 0$ and the assertion is immediate. Otherwise, after discarding the zero columns, write $z_+ = z|_{K^+}$ and set $w = (D_{\mathcal{M}}^+)^{1/2}z_+$. Entrywise domination and Hermitian symmetry give

$$|\langle R_{\mathcal{M}}w, w \rangle| \leq |w|^\top P |w| \leq \rho(P) \|w\|_2^2.$$

Hence

$$\|\mathcal{M}z\|^2 = \langle G_{\mathcal{M}}z, z \rangle = \langle G_{\mathcal{M}}^+z_+, z_+ \rangle = \|w\|_2^2 + \langle R_{\mathcal{M}}w, w \rangle \leq (1 + \rho(P)) \|w\|_2^2,$$

and $\|w\|_2^2 = \langle D_{\mathcal{M}}z, z \rangle$, which proves (22). $\square$

Theorem 3.3 (Coefficient-specific completion survival). For $z \in \mathbb{C}^K$ with $\mathcal{A}z \neq 0$, define

$$\Xi(z) := \hat{\kappa} \frac{\langle D_{\mathcal{M}}z, z \rangle}{\|\mathcal{A}z\|^2}. \quad (23)$$

Then

$$\boxed{\left(1 - \sqrt{\Xi(z)}\right)_+^2 \leq \frac{\|\mathcal{B}z\|^2}{\|\mathcal{A}z\|^2} \leq \left(1 + \sqrt{\Xi(z)}\right)^2.} \quad (24)$$

Proof. By lemma 3.2,

$$\frac{\|\mathcal{M}z\|}{\|\mathcal{A}z\|} \leq \sqrt{\Xi(z)}.$$

The triangle and reverse-triangle inequalities yield

$$(\|\mathcal{A}z\| - \|\mathcal{M}z\|)_+ \leq \|\mathcal{A}z + \mathcal{M}z\| \leq \|\mathcal{A}z\| + \|\mathcal{M}z\|.$$

Divide by $\|\mathcal{A}z\|$ and square. $\square$

Remark 3.4 (What the theorem does not use). Theorem 3.3 does not infer the phases of the missing entries from their support. The map $\mathcal{M}$, and therefore its Gram, is literal. The matrix P only majorizes the already constructed normalized Gram; it is not a replacement for that construction.

Remark 3.5 (The nontrivial regime). The lower bound in (24) is positive only if $\Xi(z) < 1$. A finite completion certificate with $\Xi(z) \geq 1$ can still be exact and useful as a stop certificate, but it gives no energy survival for that coefficient.

Corollary 3.6 (Uniform coefficient class). *Let $E \subseteq \mathbb{C}^K$ be a verified coefficient subspace on which*

$$\langle D_{\mathcal{M}}z, z \rangle \leq \chi \| \mathcal{A}z \|^2 \quad (z \in E)$$

for some $\chi \geq 0$. Then

$$\| \mathcal{B}z \|^2 \geq \left(1 - \sqrt{\widehat{\kappa}\chi}\right)_+^2 \| \mathcal{A}z \|^2 \quad (z \in E).$$

Proof. If $\mathcal{A}z = 0$, the hypothesis gives $\langle D_{\mathcal{M}}z, z \rangle = 0$. By lemma 3.2, $\mathcal{M}z = 0$, and hence $\mathcal{B}z = 0$, so the claim is trivial. If $\mathcal{A}z \neq 0$, apply theorem 3.3 with $\Xi(z) \leq \widehat{\kappa}\chi$. $\square$

4 A conservative uniform reassembly certificate

The coefficient-specific ratio is the sharpest statement available from the triangle inequality. For an archive it is often convenient to compare every term with the same global raw normalization.

Theorem 4.1 (Raw-normalized reassembly). *Let $D_{\text{raw}} > 0$, $\Gamma_{\text{pre}} > 0$, $0 \leq \eta < 1$, and $\chi \geq 0$. Suppose a literal coefficient z satisfies*

$$\| \mathcal{A}z \|^2 \geq (1 - \eta) \Gamma_{\text{pre}} D_{\text{raw}}, \quad (25)$$

$$\langle D_{\mathcal{M}}z, z \rangle \leq \chi \Gamma_{\text{pre}} D_{\text{raw}}. \quad (26)$$

Then

$$\| \mathcal{B}z \|^2 \geq \delta_{\text{sharp}} \Gamma_{\text{pre}} D_{\text{raw}}, \quad \delta_{\text{sharp}} := \left(\sqrt{1 - \eta} - \sqrt{\widehat{\kappa}\chi} \right)_+^2. \quad (27)$$

In particular,

$$\boxed{\| \mathcal{B}z \|^2 \geq \delta_{\text{reasm}} \Gamma_{\text{pre}} D_{\text{raw}}, \quad \delta_{\text{reasm}} := \left[1 - 2\sqrt{\widehat{\kappa}\chi} - \widehat{\kappa}\chi - \eta \right]_+}. \quad (28)$$

Proof. Lemma 3.2 and (26) imply

$$\| \mathcal{M}z \| \leq \sqrt{\widehat{\kappa}\chi} \Gamma_{\text{pre}} D_{\text{raw}}.$$

Together with (25), the reverse-triangle inequality gives (27).

It remains to compare the displayed constants. If the bracket in (28) is nonpositive, the claim follows from nonnegativity. Otherwise $\sqrt{1 - \eta} > \sqrt{\widehat{\kappa}\chi}$, and expansion gives

$$\begin{aligned} \delta_{\text{sharp}} &= 1 - \eta - 2\sqrt{(1 - \eta)\widehat{\kappa}\chi} + \widehat{\kappa}\chi \\ &\geq 1 - \eta - 2\sqrt{\widehat{\kappa}\chi} - \widehat{\kappa}\chi = \delta_{\text{reasm}}. \end{aligned}$$

$\square$

Remark 4.2 (Why retain the weaker constant). The sharper constant in (27) should be used when all quantities are exact. The conservative form (28) separates the represented defect η , the cross-term cost $2\sqrt{\widehat{\kappa}\chi}$, and an explicit missing self-energy reserve $\widehat{\kappa}\chi$. It is therefore convenient for a route ledger, but it is not advertised as sharp.

Proposition 4.3 (Certificate stop condition). *The data $\rho(P) < \infty$ and $\Gamma_{\text{pre}} > 0$ alone do not imply $\delta_{\text{reasm}} > 0$. A positive conclusion requires a quantitative relation among $\widehat{\kappa}$, χ , and η . If*

$$1 - 2\sqrt{\widehat{\kappa}\chi} - \widehat{\kappa}\chi - \eta \leq 0,$$

then (28) supplies no positive survival estimate and this certificate branch must stop. This failure is not a proof that the actual reassembled energy vanishes.

Proof. The formula itself gives the second assertion. For the first, take $\mathcal{M} = -\mathcal{A}$ on the one-dimensional span of z . Then $\mathcal{B}z = 0$, although both maps have finite Gram matrices and Γ_{pre} may be positive. $\square$

The one-dimensional example also shows why a claim of “the missing part is constructible” is weaker than an energy statement. Exact constructibility can produce exact destructive interference.

5 The determinant dictionary and bin-energy survival

Even a positive pre-bin energy $\|\mathcal{B}_X z_X\|^2$ need not survive determinant aggregation. The correct additional statistic is a lower modulus of the literal binning map.

Definition 5.1 (Verified reassembled subspace). Let $E_X \subseteq \mathbb{C}^{\mathcal{Y}_X}$ be a subspace constructed from literal constraints and satisfying

$$\mathcal{B}_X z_X \in E_X.$$

Its determinant-binning lower modulus is

$$s_X = s(\mathcal{J}_X; E_X) := \inf_{\substack{u \in E_X \\ u \neq 0}} \frac{\|\mathcal{J}_X u\|_2}{\|u\|_2}. \quad (29)$$

If $E_X = \{0\}$, set $s_X = 1$.

The convention for the zero subspace is immaterial because then the reassembled vector is zero. For a nonzero subspace, $s_X > 0$ is equivalent to injectivity of $\mathcal{J}_X|_{E_X}$. It is not implied by surjectivity of $\mathcal{J}_X$ on the ambient determinant space.

Theorem 5.2 (Binning–dictionary lower bound). Assume the packet of [definition 2.1](#), and suppose that for some $\Gamma_{\text{pre},X} > 0$ and $\delta_{\text{reasm},X} \geq 0$,

$$\|\mathcal{B}_X z_X\|_2^2 \geq \delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X}. \quad (30)$$

Let E_X be as in [definition 5.1](#). Then

$$D_X \geq \left(s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}} - \frac{\|r_X\|_2}{\sqrt{D_{\text{raw},X}}} \right)_+^2 D_{\text{raw},X}. \quad (31)$$

Proof. By the definition of s_X and (30),

$$\|\mathcal{J}_X \mathcal{B}_X z_X\|_2 \geq s_X \|\mathcal{B}_X z_X\|_2 \geq s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X}}.$$

Since $a_X = \mathcal{J}_X \mathcal{B}_X z_X + r_X$, the reverse-triangle inequality gives

$$\|a_X\|_2 \geq \left(s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X}} - \|r_X\|_2 \right)_+.$$

Square and use $D_X = \|a_X\|_2^2$. □

Corollary 5.3 (Exact dictionary). If $r_X = 0$, then

$$D_X \geq s_X^2 \delta_{\text{reasm},X} \Gamma_{\text{pre},X} D_{\text{raw},X}. \quad (32)$$

Definition 5.4 (Polynomial-loss ledger). Fix nonnegative exponents $\lambda_{\text{raw}}, \lambda_s, \lambda_\delta, \lambda_\Gamma, \lambda_r$. Suppose the following lower bounds hold:

$$D_{\text{raw},X} \geq X^{-\lambda_{\text{raw}}+o(1)} Q_X^3 J_X^2, \quad (33)$$

$$s_X \geq X^{-\lambda_s+o(1)}, \quad (34)$$

$$\delta_{\text{reasm},X} \geq X^{-\lambda_\delta+o(1)}, \quad (35)$$

$$\Gamma_{\text{pre},X} \geq X^{-\lambda_\Gamma+o(1)}. \quad (36)$$

Write

$$\varepsilon_{r,X} := \frac{\|r_X\|_2}{\sqrt{D_{\text{raw},X}}}. \quad (37)$$

If, for some $0 < \tau_X \leq 1$,

$$\varepsilon_{r,X} \leq (1 - \tau_X) s_X \sqrt{\delta_{\text{reasm},X} \Gamma_{\text{pre},X}}, \quad \tau_X \geq X^{-\lambda_r+o(1)}, \quad (38)$$

then λ_r is called the dictionary-separation loss.

Corollary 5.5 (Explicit certified energy-loss exponent). *Under [definition 5.4](#),*

$$D_X \geq X^{-\lambda_D^{\text{cert}} + o(1)} Q_X^3 J_X^2, \quad \boxed{\lambda_D^{\text{cert}} = \lambda_{\text{raw}} + 2\lambda_s + \lambda_\delta + \lambda_\Gamma + 2\lambda_r.} \quad (39)$$

Proof. By (38), the positive part in (31) is at least

$$\tau_X s_X \sqrt{\delta_{\text{reasm}, X} \Gamma_{\text{pre}, X}}.$$

Insert (33)–(36), square, and collect exponents. □

Remark 5.6 (A lower modulus is arithmetic data). For the sum map (13), the ambient kernel is usually large. A useful s_X therefore requires a genuinely restricted subspace E_X , defined by identities satisfied by the actual reassembled coefficient. Choosing E_X after inspecting only the target vector, or replacing it by a numerically fitted one-dimensional span, does not prove a uniform asymptotic theorem. The constraints defining E_X and the lower-modulus proof must come from archived identities independently of the value of D_X ; the proof of s_X may not invoke the determinant-energy lower bound it is meant to establish.

Remark 5.7 (Possible arithmetic hardness). Expanding $D_X = \sum_n |A_{C_X}(n)|^2$ reintroduces pairs of literal Möbius-weighted atoms constrained to have equal determinant. A lower bound at the natural scale may therefore encode a signed two-point correlation of Chowla type. The algebra above identifies that possibility; it does not remove it. Averaged results for multiplicative correlations provide important benchmarks [4, 5], but do not by themselves establish this fixed- h_0 , determinant-fiber lower bound.

6 Zero-frequency flatness and the recorded zero-slack condition

We now combine an amplitude estimate with determinant-energy survival. The calculation is elementary, but its direction is decisive.

Theorem 6.1 (General exponent transfer). *Fix $h_0 \neq 0$. Assume*

$$Q_X = X^{\beta + o(1)}, \quad J_X = X^{1 - \beta + o(1)} \quad (40)$$

for a fixed $0 < \beta < 1$. Suppose that for fixed $\eta_Z, \lambda_D \geq 0$,

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2, \quad (41)$$

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2. \quad (42)$$

Then

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{3\beta - 2 + \lambda_D - 2\eta_Z + o(1)}}. \quad (43)$$

Proof. The lower bound (42) is positive for all sufficiently large X , so $D_X > 0$ there. Hence

$$\begin{aligned} \mathfrak{F}_{h_0}(X) &= \frac{|Z_X|^2}{D_X} \\ &\leq X^{\lambda_D - 2\eta_Z + o(1)} \frac{Q_X^4}{Q_X^3 J_X^2} \\ &= X^{\lambda_D - 2\eta_Z + o(1)} \frac{Q_X}{J_X^2}. \end{aligned}$$

By (40), $Q_X/J_X^2 = X^{\beta - 2(1 - \beta) + o(1)} = X^{3\beta - 2 + o(1)}$. □

Corollary 6.2 (The high-row zero-slack transfer). *At*

$$\beta = \frac{267}{400}, \quad Q_X = X^{267/400 + o(1)}, \quad J_X = X^{133/400 + o(1)}, \quad (44)$$

the conclusion is

$$\boxed{\mathfrak{F}_{h_0}(X) \leq X^{1/400 + \lambda_D - 2\eta_Z + o(1)}}. \quad (45)$$

Consequently, the two recorded exponent bounds imply the TPC-32 hypothesis $\mathfrak{F}_{h_0}(X) \leq X^{1/400 + o(1)}$ whenever

$$\boxed{2\eta_Z \geq \lambda_D}. \quad (46)$$

Strict inequality gives a fixed power of robustness; equality gives no spare fixed-power margin. The condition is sharp for a deduction from the recorded bounds alone; it is not asserted to be necessary for the physical packet unless the two exponents are optimal.

Proof. The identity

$$3 \cdot \frac{267}{400} - 2 = \frac{801 - 800}{400} = \frac{1}{400}$$

and [theorem 6.1](#) give (45). Its recorded exponent is at most $1/400 + o(1)$ under (46). If that condition fails, these bounds alone do not close the threshold. $\square$

Corollary 6.3 (The current amplitude target requires natural energy). *The TPC-33 amplitude target*

$$|Z_X| \leq X^{o(1)} Q_X^2$$

has $\eta_Z = 0$. Under that target, (46) permits no positive certified energy loss. The sufficient zero-loss requirement is

$$\lambda_D = 0, \quad \text{equivalently} \quad D_X \geq X^{-o(1)} Q_X^3 J_X^2. \quad (47)$$

Thus any positive fixed-power determinant-energy loss cannot close this transfer from the current amplitude bound unless a new zero-mode saving compensates for it.

Proof. For the sufficient ledger condition, both exponents are nonnegative, so $0 \geq \lambda_D$ implies $\lambda_D = 0$. $\square$

Remark 6.4 (Additional losses). If a later comparison contributes an additional squared-energy loss $\lambda_{\text{aux}} \geq 0$, then the condition for this augmented certificate becomes

$$2\eta_Z \geq \lambda_D + \lambda_{\text{aux}}.$$

Such a loss cannot be hidden in an $o(1)$ term when it is a fixed positive power.

Remark 6.5 (Two distinct appearances of $1/400$). The exponent $1/400$ in (45) is the geometric baseline Q_X/J_X^2 and is the allowed TPC-32 flatness endpoint. It is not a free account from which unrecorded losses may be spent. Separately, whenever the wider route compares accumulated fixed-power losses with an inherited $1/400$ margin, the operational rule is strict:

$$\text{continue only if total loss} < \frac{1}{400}; \quad \text{stop if total loss} \geq \frac{1}{400}.$$

In particular equality in (46) is a valid conditional implication but has zero robustness against any further fixed-power degradation.

7 Sharp nonidentifiability and no-go models

The next examples show that each interface in [section 1](#) carries information absent from the preceding one.

Proposition 7.1 (Unsigned completion data do not determine the zero mode). *For every integer $N \geq 2$, there are vectors $u, v \in \mathbb{C}^N$ with identical support, identical coordinate moduli, and identical energy, but*

$$\left| \sum_{j=0}^{N-1} u_j \right| = N, \quad \sum_{j=0}^{N-1} v_j = 0. \quad (48)$$

Consequently, no statistic depending only on unsigned support, coordinate masses, or total energy can determine zero-frequency flatness.

Proof. Take $u_j = 1$ and

$$v_j = \exp(2\pi i j/N).$$

Both vectors have full support, $|u_j| = |v_j| = 1$, and squared norm N . The first sum is N , while the second is the sum of all N -th roots of unity and vanishes. $\square$

Remark 7.2 (Real version). For even N , the real vectors

$$u = (1, \dots, 1), \quad v = (1, -1, 1, -1, \dots, 1, -1)$$

already give the same conclusion. Thus the obstruction is not caused by complex phases.

Proposition 7.3 (Raw atomic mass need not survive a determinant bin). *There is no positive universal constant c such that*

$$\|\mathcal{J}w\|_2^2 \geq c\|w\|_2^2$$

for every sum-binning map and every raw vector w . Indeed, two nonzero raw atoms in a single determinant bin can have raw energy 2 and determinant energy 0.

Proof. Let $\mathcal{Y} = \{y_1, y_2\}$, let both keys have the same determinant label n_0 , take unit multipliers, and set

$$w(y_1) = 1, \quad w(y_2) = -1.$$

Then $\|w\|_2^2 = 2$, while

$$(\mathcal{J}w)(n_0) = 1 - 1 = 0.$$

Thus $w \in \text{Ker } \mathcal{J}$, and the ambient lower modulus is zero. $\square$

Corollary 7.4 (Support survival is not energy survival). *The facts that every raw atom is represented, every determinant bin is nonempty, and the raw energy has its natural order do not imply a positive lower bound for D_X .*

Proof. [Proposition 7.3](#) has all three properties and post-bin energy zero. $\square$

Example 7.5 (The same majorant can hide opposite reassemblies). *In one dimension take $\mathcal{A}z = z$. The choices $\mathcal{M}_+z = z$ and $\mathcal{M}_-z = -z$ have the same missing Gram, the same diagonal mass, and the same Perron constant $\hat{\kappa} = 1$. Yet*

$$(\mathcal{A} + \mathcal{M}_+)z = 2z, \quad (\mathcal{A} + \mathcal{M}_-)z = 0.$$

The coefficient-specific bound correctly permits both outcomes when $\Xi(z) = 1$. Therefore the Perron majorant is a bound, not a phase reconstruction theorem.

Proposition 7.6 (A small zero mode does not force natural energy). *For every $\varepsilon > 0$ there is a nonzero coefficient a with $\sum_n a_n = 0$ and $0 < \|a\|_2^2 < \varepsilon$.*

Proof. Take $a = (t, -t)$ with $0 < 2|t|^2 < \varepsilon$. $\square$

This last example is elementary but prevents a common reversal: cancellation in Z_X does not furnish a lower bound for D_X . The numerator and denominator in $\mathfrak{F}_{h_0}(X)$ require separate arithmetic inputs.

Remark 7.7 (Averaging over shifts). An averaged- h lower bound for determinant energy is compatible with a zero or tiny value at one prescribed h_0 . Hence mean-square estimates over shifts cannot be relabeled as the fixed- h_0 hypothesis (42). This is a quantifier obstruction, not a technical normalization issue.

8 Conditional transfer to the matched shell and full block

We now record the strongest valid arithmetic implication. Every premise is explicit.

Theorem 8.1 (Literal fixed-shift flatness transfer). *Fix $h_0 \neq 0$, and let a_X be the actual determinant coefficient of the TPC-32 matched packet with*

$$C_X = \lfloor J_X \rfloor, \quad Q_X J_X \asymp X, \quad Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}.$$

Write

$$N_{0,X} := J_X Q_X^2 \asymp X Q_X \tag{49}$$

for the literal raw scale of this matched shell. Suppose there are fixed exponents $\eta_Z, \lambda_D \geq 0$ for which the following assumptions hold:

- (T1) the complete packet of [definition 2.1](#) is literal, with the same signs, units, orientation, and global raw normalization as a_X ;
- (T2) the missing-Gram hypotheses of [theorem 4.1](#) give $\delta_{\text{reasm},X}$;
- (T3) an exact determinant map $\mathcal{J}_X$, a verified subspace E_X , and a dictionary residual r_X satisfy [theorem 5.2](#);
- (T4) the resulting lower bound proves

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2;$$

- (T5) independently, the actual zero mode satisfies

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2$$

with $2\eta_Z \geq \lambda_D$.

Then

$$\mathfrak{F}_{h_0}(X) \leq X^{1/400+o(1)}. \tag{50}$$

Consequently, by the conditional zero-frequency theorem of TPC-32, every fixed

$$\sigma < \frac{133}{400} \tag{51}$$

is a power-saving exponent relative to its raw scale $N_{0,X}$ for the complete matched cutoff shell. Equivalently, if $\mathcal{S}_X$ denotes that shell, then

$$|\mathcal{S}_X| \leq X^{-\sigma+o(1)} N_{0,X}. \tag{52}$$

Proof. Assumptions (T4)–(T5) and [corollary 6.2](#) give (50). Assumption (T1) identifies this ratio with the distinguished coefficient used in TPC-32, rather than with a surrogate. Its conditional high-row closure theorem then gives (52) for every σ satisfying (51) [7]. $\square$

Remark 8.2 (What completion contributes). Assumptions (T2)–(T3) can prove (T4), but they do not prove (T5). In particular, a complete carrier census, a positive reassembly constant, and an injective determinant dictionary still do not cancel the signed zero mode. The amplitude estimate remains an arithmetic premise.

The matched shell is not automatically the whole Type-II residual. The global step requires an exact block identity of the kind isolated in TPC-18 [6].

Theorem 8.3 (Conditional full-block transfer). *Fix $h_0 \neq 0$. Assume that the physical hard remainder has an exact decomposition*

$$\mathcal{B}_{h_0,\delta}(X) = \sum_{b \in \mathcal{B}_X} w_{b,X} \mathcal{S}_{b,X} + R_X, \quad |\mathcal{B}_X| = O(\log X), \tag{53}$$

where:

- (F1) every $\mathcal{S}_{b,X}$ is a literal matched packet with complete native keys and the same fixed shift h_0 , and $N_{0,b,X}$ denotes its own literal raw scale;
- (F2) the hypotheses of [theorem 8.1](#) hold uniformly in b , with its own literal coefficient $a_{b,X}$, determinant dictionary, and certificate data for every block;
- (F3) $|w_{b,X}| \leq X^{o(1)}$, $R_X = o(X)$, and for some fixed $0 < \sigma_* < 133/400$,

$$\sum_{b \in \mathcal{B}_X} |w_{b,X}| N_{0,b,X} X^{-\sigma_*} = o(X); \quad (54)$$

- (F4) all calibration, drift, endpoint, nonprimitive, and prime-power terms omitted from (53) are already included in R_X with their literal signs.

Then

$$\mathcal{B}_{h_0,\delta}(X) = o(X). \quad (55)$$

Proof. Choose a fixed σ with $\sigma_* < \sigma < 133/400$. Uniform application of [theorem 8.1](#) gives

$$|\mathcal{S}_{b,X}| \leq X^{-\sigma+o(1)} N_{0,b,X}.$$

The fixed gap $\sigma - \sigma_*$ absorbs the uniform $o(1)$. Insert the resulting $|\mathcal{S}_{b,X}| \leq X^{-\sigma_*} N_{0,b,X}$ in (53), sum absolutely, and use (54) and $R_X = o(X)$. $\square$

Remark 8.4 (Two different uses of $\mathcal{B}$). The synthesis operator $\mathcal{B}_X = \mathcal{A}_X + \mathcal{M}_X$ acts on completion coefficients. The scalar $\mathcal{B}_{h_0,\delta}(X)$ in (53) is the physical hard remainder. The shared letter does not identify these differently typed objects.

Remark 8.5 (Status of the cover). [Theorem 8.3](#) is a logical implication, not a proof of (53). The all-block cover was isolated earlier as an additional hypothesis. Nor does the present paper verify the uniform zero-mode or energy premises for even one growing physical block. Therefore (55) is not an unconditional result.

Remark 8.6 (No prime-pair claim). For an admissible even h_0 with nonzero main-term integral $\mathcal{I}_0(W)$, in the earlier Vaughan normal form a theorem of the strength $\mathcal{B}_{h_0,\delta}(X) = o(X)$, together with all inherited identities, would reach a weighted fixed-shift Hardy–Littlewood asymptotic. That observation measures the strength of the missing premises. It does not prove them, and it must not be restated as a twin-prime or parity result. For the standard analytic-number-theory framework in which such reductions sit, see Iwaniec and Kowalski [3].

9 Reproducible audit, claim boundary, and next gate

The transfer is useful only if every object can be checked against the literal packet. We record a minimum archive.

9.1 Carrier and dictionary census

For each audited scale X , archive:

- (A1) the fixed value h_0 , the exact Q_X, J_X, C_X , and the version of every upstream identity;
- (A2) the complete ordered native key sets $\mathcal{K}_X, \mathcal{Y}_X, \mathcal{N}_X$, including field names, integer ranges, orientations, contents, provenance labels, and cryptographic hashes;
- (A3) the exact sparse arrays of $\mathcal{A}_X, \mathcal{M}_X, \mathcal{B}_X$, before thresholding, together with dimensions, ranks, and actual nonzero supports;
- (A4) the literal coefficient z_X , the actual determinant coefficient a_X , and a direct exact check of $r_X = a_X - \mathcal{J}_X \mathcal{B}_X z_X$;

- (A5) the determinant label ν_X , every multiplier $\tau_X(y)$, every fiber cardinality, and the sparse matrix of $\mathcal{J}_X$;
- (A6) the single global $D_{\text{raw},X}$, with its atomic measure and units; no block-renormalized replacement;
- (A7) $G_{\mathcal{M},X}$, its exact active diagonal $D_{\mathcal{M},X}$, the normalized residual $R_{\mathcal{M},X}$, the majorant P_X , an entrywise proof of $|R_{\mathcal{M},X}| \leq P_X$, and a certified bound for $\rho(P_X)$;
- (A8) the quantities $\eta_X, \chi_X, \Gamma_{\text{pre},X}, \Xi_X(z_X), \delta_{\text{sharp},X}$, and $\delta_{\text{reasm},X}$;
- (A9) the defining constraints for E_X , an exact proof that $\mathcal{B}_X z_X \in E_X$, and a certified lower bound for s_X ;
- (A10) the exact values $Z_X, D_X, \mathfrak{D}_{h_0}(X)$, and $\mathfrak{F}_{h_0}(X)$, followed by the complete polynomial-loss ledger of [corollary 5.5](#).

Exact integer, rational, algebraic, or interval certificates should be used whenever feasible. Floating-point tolerances are reported separately and never used to change the exact support. A regression over finitely many X may motivate an exponent, but it does not prove the asymptotic inequalities in [theorem 8.1](#).

9.2 Six mandatory physical checks

Before any L1 claim, answer all six questions:

- (C1) Are the coefficients literal physical coefficients rather than majorant, trace, or averaged-family coefficients?
- (C2) Is the same prescribed nonzero h_0 retained in every map?
- (C3) Is the atomic normalization global and physical?
- (C4) Is the representation canonical, or has every multiplicity in a nonminimal representation been restored?
- (C5) Is the support the actual exact post-aggregation support?
- (C6) Does the endpoint ledger keep every fixed-power loss explicit and obey the strict 1/400 stop rule?

Failure of any check returns the result to L0.

9.3 Decision protocol

The next arithmetic decision is a deliberate trichotomy.

Route A. Construct the literal dictionary with $r_X = 0$, and prove

$$s_X, \delta_{\text{reasm},X}, \Gamma_{\text{pre},X} \geq X^{-o(1)}, \quad D_{\text{raw},X} \geq X^{-o(1)} Q_X^3 J_X^2.$$

Then [\(47\)](#) follows and the remaining independent task is the zero-mode amplitude estimate.

Route B. Prove an amplitude gain $\eta_Z > 0$. Then a determinant-energy loss $\lambda_D \leq 2\eta_Z$ is admissible, subject to the separate strict endpoint ledger.

Route C. Show that the required lower bound for D_X , after exact expansion, contains an unresolved fixed-shift signed correlation of the same strength as a Chowla/parity input. Then record a negative transfer theorem and stop this route rather than rename the missing arithmetic as a completion lemma.

9.4 Claim table

Statement	Status here	Missing physical input
Missing-Gram energy bound	Exact L0 theorem	Literal $\mathcal{M}_X$ and asymptotic certificate statistics
Uniform reassembly	Exact conditional L0 theorem	Positive $\delta_{\text{reasm},X}$ at physical scale
Determinant-energy lower bound	Exact conditional L1 implication	Literal $\mathcal{J}_X$, lower modulus, and small r_X
Condition $2\eta_Z \geq \lambda_D$	Exact criterion for this exponent transfer	Bounds for the actual numerator and denominator
Matched-shell saving	Conditional inherited implication	All hypotheses of theorem 8.1
$\mathcal{B}_{h_0,\delta}(X) = o(X)$	Conditional full-block implication	Exact cover, normalization sum, and uniform packet estimates
Fixed-shift prime-pair asymptotic or twin primes	Not proved	L2 arithmetic closure and all inherited bridges

9.5 Conclusion

The new point is not a hidden cancellation theorem. It is an exact factorization of the remaining inference:

literal completion
 $\Downarrow$ Gram survival
pre-bin energy
 $\Downarrow$ restricted binning + dictionary
determinant energy
 $\Downarrow$ independent zero-mode estimate
zero-frequency flatness.

The first two downward arrows are now governed by explicit, falsifiable constants. The final arithmetic inputs remain open. That separation prevents exact finite algebra from being mislabeled as fixed-shift parity progress while leaving a precise route for a genuine future advance.

References

- [1] D. R. Heath-Brown. A parity problem from sieve theory. *Mathematika*, 29(1):1–6, 1982. doi: 10.1112/S0025579300012109.
- [2] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013. doi: 10.1017/CBO9781139020411.
- [3] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 2004. doi: 10.1090/coll/053.
- [4] Kaisa Matomäki, Maksym Radziwiłł, and Terence Tao. An averaged form of Chowla’s conjecture. *Algebra & Number Theory*, 9(9):2167–2196, 2015. doi: 10.2140/ant.2015.9.2167.
- [5] Terence Tao. The logarithmically averaged Chowla and Elliott conjectures for two-point correlations. *Forum of Mathematics, Pi*, 4:e8, 2016. doi: 10.1017/fmp.2016.6.
- [6] Liang Wang. Two quadraticizations of a Möbius tail: LCM cutoff return, generic row determinants, and nonprimitive endpoint collapse, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-18-mobius-tail-determinant-dispersion. Preprint.
- [7] Liang Wang. Matched prime–Möbius cutoff shells: Rank–two factorization, determinant–frequency dispersion, and the distinguished zero–frequency gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-32-matched-cutoff-frequency-gate. Preprint.

- [8] Liang Wang. Primitive affine columns at a distinguished zero frequency: Fixed-determinant projective plancherel, structured-mask disintegration, and the q^3 energy gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-33-primitive-affine-columns. Preprint.
- [9] Liang Wang. Primal-dual completion certificates for fixed-shift primitive entry kernels: Exact quotient gauges, obstruction witnesses, and perron aggregation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-75-primal-dual-completion-certificates. Preprint.
- [10] Liang Wang. Exact sparse incidence decomposition for provenance completion programs: Componentwise primal-dual certificates and verified carrier census protocols, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-77-sparse-incidence-completion-certificates. Preprint.
- [11] Liang Wang. Scalar hermitian perron saturation and cycle-holonomy defects: Exact equality, frustration energies, and weighted bounds, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-80-scalar-hermitian-perron-holonomy. Preprint.